		Target engagement	Dose	Efficacy	Safety
#32	**Hodgkins lymphoma** anti-PDCD1 antibodies (6)	**2014** Ex-vivo PD-1 binding on activated human T cells PD-1 occupancy >70% in patients	**2015** Mouse syngeneic tumour PK-PD (MC38, Sa1/N) with ex-vivo… Predicted active dose was reached	**2020** cHL cell lines and microdissected Reed-Sternberg cells (9p24.1/PD-L1/PD-L2 genomics)	**2016** Mouse allogeneic bone-marrow transplant GVHD… △ Silent on immune toxicity seen i…
	Hodgkins lymphoma anti-CD274 antibodies (3)	**2022** In-vitro PD-L1 binding and PD-1 blockade assays (engineered PD-L1/PD-1 cell lines, CD4+ T-cell…	**2022** Cynomolgus monkey repeat-dose toxicology (4- and 26-week GLP studies, NOAEL 200…	**2022** 9p24.1/PD-L1 genetics of Reed-Sternberg cells (patient… Patient result unclear	no model relied on
#33	**open-angle glaucoma** ROCK inhibitors (6)	**2017** Living mouse eye, OCT of conventional outflow tissue, topical… Human facility up 22%	**2013** Normotensive monkey eye, topical netarsudil 0.04%	**2026** ARHGEF12 rs11824032 risk genotype (African-ancestry POAG… Genotype did not predict response	**2017** Monkey 9-month topical ocular toxicology
	open-angle glaucoma PTGER2 agonists (3)	**2015** Cynomolgus monkeys with laser-induced ocular hypertension… Patient IOP fell as in monkeys	**2018** Cynomolgus monkeys with laser-induced ocular hypertension… Once-daily regimen held in patients	**2018** Cynomolgus monkeys with… Phase 3 IOP noninferior to latanoprost	**2022** Cynomolgus monkey 13- an… Corneal thickening seen in patients too
#34	**age-related macular…** anti-VEGFA antibodies (5)	**2019** Rat subretinal/suprachoroidal RGX-314 with VEGF-induced… Patients expressed RGX-314 protein	**2020** Rabbit and dog intravitreal toxicology (abicipar) IOI 15% below stated margin	**2019** Rat subretinal/suprachoroidal RGX-314 with VEGF-induced… Patient result unclear	**2019** NZW rabbit intravitreal TK/tolerability (brolucizumab) △ Intraocular inflammation missed…
Allergy	**allergic disease** allergen (immunotherapy) (5)	**2019** Patient PBMC Fel d 1-specific CD4+ T cells (ex vivo), Cat-PAD	**2016** Allergen exposure chamber	**2018** Allergen exposure chamber Posted results: CS 1.04/1.00 vs 1.05	**2016** Patient blood basophils, histamine release (ex vivo)
	allergic rhinitis PTGDR2 antagonists (4)	**2013** Human whole-blood PGD2 eosinophil shape-change assay	**2017** Asthmatic allergen challenge	**2012** OVA mouse rhinitis Mouse nasal-symptom relief absent in Phase 3	**2017** Asthmatic allergen challenge
	allergic rhinitis allergen (immunotherapy) (8)	**2017** Conjunctival test, gp-ASIT+	**2018** Allergen exposure chamber	**2017** Allergen exposure chamber △ Confirmatory field trial missed…	**2018** Allergen exposure chamber
Autoimmune	**Sjogren syndrome** BTK inhibitors (3)	**2021** Rat BTK occupancy and collagen-induced arthritis… Human blood BTK occupancy >95%	**2021** Semi-mechanistic PK/PD occupancy model (tirabrutinib) Not measured in patients	**2022** Rat BTK occupancy and collagen-induced arthritis (remibrutinib)	**2016** Rat and dog repeat-dose toxicology (remibrutinib) Model data unpublished
	Sjogren syndrome PIK3CD inhibitors (3)	**2019** AdV5 cannulation focal sialadenitis mouse (seletalisib) Patient gland foci and signalling fell	**2015** AdV5 cannulation focal sialadenitis mouse (seletalisib) No checkable prediction	**2019** Spontaneous NOD.ShiLtJ and NOD.B10 Sjögren's mice… No checkable prediction	**2016** Leniolisib rat and monkey repeat-dose toxicology Model data unpublished
	psoriasis JAK1 inhibitors (3)	**2018** Human whole-blood… JAK PD biomarkers moved, already known	**2022** Rat, monkey repeat-dose tox Monkey NOAEL margin held in patients	**2018** Rat adjuvant-induced arthriti… Arthritis model post-dated JAK1 psoriasis wins	**2014** Rat, monkey repeat-dose tox
	psoriasis PDE4 inhibitors (3)	**2020** Human Th17 ex vivo skin model Not measured in patients	**2010** Mouse phorbol ear edema	**2022** Imiquimod psoriasis-like mouse (topical DRM02), with phorbol-ester ear and Th2…	**2011** Rabbit primary skin irritation and mouse local lymph node assay (2% ointment)
	psoriasis RORC inverse agonists (4)	**2015** Human whole-blood IL-17 secretion assay △ No IL-17 change in plaques	**2015** Human whole-blood IL-17 secretion assay	**2017** Imiquimod mouse, topical GSK2981278 Patient result unclear	**2015** Nonclinical toxicology species (not public) Model data unpublished
	psoriasis anti-IL17A antibodies (4)	**2012** Human keratinocytes stimulated with IL-17 and TNF	**2016** Cynomolgus repeat-dose… Cyno NOAEL margin held at clinical dose	**2023** Mice injected with human IL-17A, plasma KC (CXCL1) readout PASI 90 71% vs 0% placebo	**2016** People with inborn errors of IL-17 immunity
	psoriasis anti-IL23A antibodies (3)	**2015** Human psoriasis skin xenograft on AGR129 mice (anti-p19 clone 7G10)	**2018** Cynomolgus monkey repeat-dose toxicology Cyno NOAEL margin held in patients	**2018** Human psoriasis skin xenograft on AGR129 mice (anti-p19 clone 7G10)	**2018** Mouse intracellular-infection… Mouse infection models: no excess risk, held
	systemic lupus erythematosus BTK inhibitors (3)	**2021** Human PK/PD model of BTK inhibition fitted to fenebrutinib… △ Evobrutinib: anti-dsDNA…	**2021** Human PK/PD model of BTK inhibition fitted to fenebrutinib healthy-volunteer data	**2023** NZB/W F1 lupus mice (spontaneous or IFNα-accelerated… Elsubrutinib arm stopped for futility	**2017** Nonclinical GLP toxicology of member drugs (species not public) Model data unpublished
	ulcerative colitis S1PR1 agonists (3)	**2016** Mouse blood lymphocyte… Patient lymphocytes fell 49%, mice 51%	**2023** Rat and dog repeat-dose toxicology (etrasimod)	**2019** TNBS and CD4+CD45RBhi transfer colitis mice	**2023** hERG current assay in vitro (etrasimod) △ Dogs silent; patients' heart rate fell
Endocrine	**endometriosis** GNRHR antagonists (3)	**2022** Rat HPG axis (LH in male rats; E2 in pro-oestrus female rats)… Rat E2 suppression seen in women	**2023** Rat surgically induced (autotransplant) endometriosis,… Not measured in patients	**2024** Patient-derived endometrioma xenograft 'murine avatars' in NOD/SCID mice (lesion size by…	**2018** Rat surgically induced… Rat said no bone loss; women lost BMD
	hypopituitarism GHR agonists (3)	**2012** Juvenile cynomolgus monkey (IGF-1 PK/PD)	**2014** Juvenile cynomolgus monkey (IGF-1 PK/PD) Children needed 3.6x monkey dose	**2015** Hypophysectomized rat (weight gain, IGF-1) △ Somavaratan inferior to daily GH	**2012** Pig and monkey local tolerability (lipoatrophy)
	pituitary dwarfism GHR agonists (3)	**2018** Somapacitan population… IGF-I rose dose-dependently in REAL 3	**2020** Somapacitan population PK/PD (IGF-I indirect-response) model from phase 1 data, simulati…	**2017** Hypophysectomized rat growth assay	**2015** Monkey and pig toxicology (VRS-317)
	secondary hyperparathyroidism CASR positive modulators (4)	**2020** Patient-derived primary parathyroid cells ex vivo (primary HPT adenoma tissue)	**2014** Population PK/PD model of etelcalcetide (PTH-calcium homeostasis, CaSR cooperativity)	**2016** Population PK/PD model of… Predicted ~75% responders; 74-75% observed	**2013** Rat PD and rat/dog toxicology
	type 1 diabetes mellitus SLC5A2 inhibitors (4)	**2014** STZ-diabetic rat	**2008** STZ-diabetic rat No checkable prediction	**2019** STZ-diabetic rat	**2018** Diet-induced obese rat, insulin-replete, dapagliflozin, 38 days △ Dapagliflozin DKA, unflagged by…
Gastroenterology	**exocrine pancreatic insufficiency** PNLIP replacement / gene therapy (3)	**2018** Pancreatic duct-ligated, embolized minipig (PEI model)	**2015** Pancreatic duct-ligated, embolized minipig (PEI model) No checkable prediction	**2016** Pancreatic duct-ligated dogs (Mayo Clinic, DiMagno), dosed wit… △ CFA far below PERT in CF…	**2017** Sponsor nonclinical toxicology (species not public) Model data unpublished
	GERD ATP4A blockers (3)	**2026** Semi-mechanistic multi-leve… Reproduces fexuprazan pH holding time	**2017** Stimulated gastric acid secretion in dogs No checkable prediction	**2019** Rat reflux oesophagitis vs… Rat model post-dates EE healing result	**2025** Rat repeat dosing: gastrin, emptying, microbiota Not measured in patients
	opioid-induced constipation OPRM1 antagonists (4)	**2011** Rat opioid-induced small-intestinal transit inhibition Not measured in patients	**2011** Rat GI-transit vs tail-flick analgesia dose window	**2019** Mouse morphine-inhibited intestinal transit and gastric emptyi… Phase 2b CSBMs rose over placebo	**2011** Morphine-dependent rats (precipitated withdrawal)
	primary biliary cholangitis NR1H4 agonists (3)	**2024** Rodent FXR target-gene P… Cilofexor rodent PD reproduced in phase 1	**2019** Rodent FXR target-gene PD models No checkable prediction	**2020** Mdr2-/- mice	**2020** Mdr2-/- mice Ductular-reaction prediction never tested in PBC
Hematology	**hemophilia A** F8 replacement / gene therapy (8)	**2025** Cynomolgus macaques with pre-existing anti-AAV5 antibodies given valoctocogene roxaparvovec	**2021** FVIII-knockout mice + cynomolgus macaques given AAVhu37.E03.TTR.hFVIIIco-SQ…	**2021** Hemophilia A dogs (Chapel Hill / Queen's colonies) △ Mouse-stable BAX 888 lost in man	**2023** Hemophilia A dogs (Chapel Hill / Queen's colonies) △ ALT rise in 81% of patients
	hemophilia B F9 replacement / gene therapy (7)	**2019** Cynomolgus monkeys with pre-existing anti-AAV5 NABs given AMT-060 (AAV5-hFIX)	**2017** Hemophilia B dogs, AAV8 canine FIX-Padua No checkable prediction	**2024** Juvenile hemophilia B dogs, canine surrogate vector ABR noninferior; dogs reproduced	**2015** FIX-knockout and normal mice given scAAV8.FIXR338L (BA… △ No such harm in patients
	PNH anti-C5 antibodies (4)	**2018** Hemolysis assay with C5 R885H variant	**2018** Population PK/PD (free C5, LDH) model-based simulation of Phase 1/2 data (ravulizumab)	**2018** In-vitro complement hemolysis assay in human serum	**2020** Drug–target–drug complex assay, human C5
Metabolic	**Fabry disease** GLA replacement / gene therapy (4)	**2019** Gla-knockout (alpha-Gal A deficient) Fabry mouse	**2023** Six-month IV toxicology, mice and cynomolgus monkeys… Tox margin held at 1 mg/kg	**2019** Gla-knockout (alpha-Gal A deficient) Fabry mouse	**2021** Gla-knockout Fabry mice given clinical-scale ST-920 (AAV2/… △ Patients' liver enzymes rose;…
	glycogen storage disease II GAA replacement / gene therapy (3)	**2023** Human whole-blood / neutral-pH enzyme stability assay (cipaglucosidase alfa with miglustat)	**2023** Rat and cynomolgus monkey repeat-dose toxicology (exposure margin)	**2021** Gaa knockout mice (Gaa tm1Rabn; tissue glycogen, muscle… 6MWD favoured, superiority missed	**2023** Rat and cynomolgus monkey repeat-dose toxicology
	hyperlipidemia CETP inhibitors (3)	**2014** Human CETP/apoA-I… Human CETP activity inhibited 65-91%	**2014** Human CETP/apoA-I transgenic mouse △ Exposure far below prediction	**2011** Human CETP loss-of-function genotypes ☐ Outcome trial negative	**2011** Rat blood pressure + human adrenal H295R cells
	hyperlipidemia HCAR2 agonists (3)	**2015** Rat, guinea pig plasma NEFA Authors: models predicted human NEFA response	**2012** Rat plasma FFA suppression (Merck screen) △ NEFA suppression not held	**2015** Rat, guinea pig plasma NEFA	**2016** Fasted beagle dog, plasma FFA and flushing